

DCA Monday February 4, 2019

Monday, February 04, 2019 7:52 AM

Dave kicking it off

Ft. Collins was started above

Marriott100 - conf. password

Round robin introductions

1. Dan on Data Centricity

- a. What it is; shared schema, possibly across databases
- b. Data driven vs data centric
 - i. Results become data
- c. DCA
 - i. Open standards are key; portability
 - ii. Data should be self-describing as part of the data
 - iii. Flexibility - your environment **will** change, you need to be able to adapt
 - iv. Algorithmic result persisted in the database
 - v. Architecture should define its own integrity and manage it
 - vi. Simplicity, elegance
 - 1) Dave adding his comment about number of words people know
 - a) 20K by high school, 40K end of college ???
 - 2) Model should be understandable enough to easily find things

vii. Fidelity

- 1) Contextual meaning is very hard
 - a) Touched on folksonomies
 - b) Uche - not only agreeing context, but competing contexts.
 - i) Important to capture agreements **and** disagreements.

viii. Security- robust and fine-grained

- 1) Contained in the data, NOT the application
 - a) Chris - we will have multiple owners so one identifier
 - b) Alan - some work being done with identity management
 - c) Uche - authentication has been at the heart of it. Need to get non-repudiation, blockchain, at the center. Having the ledger that certify auth of data as it is entered

ix. Provenance - need to easily surface it for consumption patterns

x. Extensibility

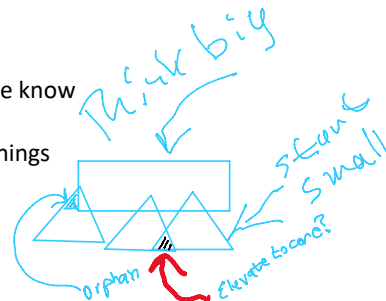
- 1) Organizations need to be able to accommodate specific needs beyond the core definition

xi. Federability

- 1) Uche - Apache Druid - Data lake vs data river serviced by micro-services.

d. DCA covers concerns using semantic tech

e. There are nonsemantic approaches to data-centricity



- i. Dave: IRM in Sweden has implemented a data centric
- f. Review of the stuff he already discussed.
 - i. Dave: App store comparison
 - ii. Uche - used app store approach at Zepheira
 - iii. DCA graphic showing layers
 - 1) Data
 - a) Data includes the metadata
 - 2) Federated queries
 - a) Single query across it all
 - 3) Core ontology
 - a) Most of the meaning is contained here, referenced by other ontologies
 - 4) Data security
 - a) Rules captured as data, logging resulting events as data too
 - 5) Data Identity management
 - a) Mdm redone
 - 6) Integrity, constraints, triggers
 - a) Generally SHACL applies here, possibly along with other things
 - 7) Programmatic behaviors
 - a) Ideally, these are microservices
 - b) Boris: changes are controlled/curated mutations of your data
 - c) Chris: making the data changes independent from each other
 - d) Uche: in this layer, declarative that is data; need to think about how to implement
 - e) Eric: how much of the metrics can you get into data
 - f) Dave: there still has to be some code here.
 - 8) Domain ontologies and taxonomies
 - a) Andrea: pre and post terminology definition for disambiguation
 - b) Dave: orphans - not in core so not an extension so the rest of the organization won't know what it means. Had it been a specialization of something in the core, there would be more clarity across the organization.
 - c) Michael: keeping things that aren't actually real, from things that are.
 - 9) Endpoint access management
 - 10) User interfaces
 - iv. Michael - does daa in motion jump out of the DCA
 - v. Eric - how do we get the current software industry to not out-marketing DCA into oblivion.

2. Dave

- a. Three things
 - i. Low code, no code
 - ii. Graph db, does it have to be RDF standards
 - iii. Semantics

-----Afternoon-----

- 1. Alan and Dave on data centric market status, case studies and outlook
 - a. Application centric it mindset; hat an IT problem? Buy a solution.
 - i. This model dis-empowers the user to understand their environment.

Sidebar from 1.d.:

- Need to have a bridge from tactical to strategic
- Dave: one of our clients is doing this.
 - GDPR was taken lightly, California now has CCPA so it is starting to be taken

1. Alan and Dave on data centric market status, case studies and outlook
 - a. Application centric it mindset; hat an IT problem? Buy a solution.
 - i. This model dis-empowers the user to understand their environment.
 - ii. Dave: there isn't any funding for what we are interested in; why is that?
 - 1) No template copy as we are different than prior approaches
 - 2) Need a better value proposition
 - b. Own the problem first, then build a solution.
 - i. Strategy and planning
 - ii. Execution
 - iii. Human in the loop (missed the company doing this)
 - 1) Photo tagging
 - iv. Data-centric design thinking (part of strategy and planning, perhaps an overlay crossing both sides)
 - 1) MVP - minimum viable product
 - 2) Dan: is this a gateway to low-code, no-code
 - v. Relationship-rich modeling (part of strategy and planning, perhaps an overlay crossing both sides)
 - vi. Data-centric infrastructure
 - vii. Model-driven development
 - viii. Human and machine learning loops
 - c. All the big four tried out Watson to see how it would work for them.
 - i. Need to think beyond end-to-end packaged software and SaaS solution.
 - 1) Cognitive computing
 - ii. Consider how to get the business and IT together building and solving foundational data problems.
 - 1) Bespoke projects with worthy objectives and practical means of success
 - 2) Dave: Salesforce came so close
 - 3) Eric: enterprise software perspective is you have to say yes to all comers.
 - a) Siebel was in upgrade gridlock.
 - 4) Alan: identity management industry introduced Sovrin Network that is now on 6 continents. Has 6 stewards such as IBM, Oracle. This uses blockchain to assure identity
2. Break
3. Chris Harding - Lacibus - vendor presentation
 - a. Virtual Data Lake
 - b. ...
 - c. Low code/no code
 - i. Not thinking this is a thing
 - d. Codeless data management?
 - i. Yes
 - e. Virtual data lake
 - i. Applications combine data from multiple sources
 - ii. Source owners control access to their data
 - f. Metadata
 - i. Open standards group has metadata for IoT data.
 - g. OD-F - Open standards group standard for metadata
 - h. <https://lacibus.com>
 - i. Free trial: chris@lacibus.net

strategic

- Dave: one of our clients is doing this.
 - GDPR was taken lightly, California now has CCPA so it is starting to be taken seriously.
- Meika: if we are going to be disruptive, we need to be able to guide our clients as to what each step on the way looks like
- Eric: what is the killer app here that will grab folks attention.
- Uche: what is customer supporting angle as that is hot - Alexa, Siri, et.
- Nationwide had CDO role a while ago, (Jim Tyo=current CDO)
- PWC added CDO role in 2015
- Boris: pitch cutting their taxes by reducing their costs
 - Silo tax!

Pedro Domingos, *The Master Algorithm*, 2015

Kbmedia

- i. Send request and he will set you up.
 - ii. Currently using LinkedIn and Google for identity management
- j. How it works:
 - i. You put in an item and it returns triples
- 4. Tatinana - Lymba
 - a. Ontology driven NLP
 - b. NLP for creating ontologies
 - c. ...
 - d. Text mining framework Text -> knowledge -> applications
 - e. Knowledge
 - i. Concepts
 - ii. Semantic relations
 - f. Example: contract processing
 - g. Customizable semantic knowledge extraction
 - h. Automated ontology creation/extension
 - i. Jaguar
 - i. Unsupervised
 - ii. Gets about 50% of the gold
 - j. K-Extractor
 - i. Ontologist supervised
 - k. Case studies
- 5. Back to the state of the DCA
 - a. Referring to Kurt Cagle's *Semantic Zoo* article on Forbes.
 - b. Blockchain
 - i. Things are
 - 1) 'on chain' - the secure, immutable component
 - 2) 'off chain' - the insecure, mutable environment i.e. the open internet from which the transactions are initiated.
 - a) Think about identity management; if it is bad off chain, that bad data gets into the on chain portion.
 - c. Dave: we need to have agreement going in for this all to work
 - d. Sovrin identity personal hub
 - i. See also Tim Burners-Lee's Solid
 - 1) Starting to see some use
 - ii. The issue is correlatable information
 - iii. Uche's brother has published a paper on personal health records using Solid
- 6. Market buzz - Alan Morrison
 - a. Data value chain
 - b. SaaS and the cloud
 - i. Enterprises use average of 1200 cloud services - 2017
 - ii. Data localization has become an issue
 - c. ...
 - d. GraphQL
 - i. Uche: REST++
 - e. Data-centric audit and protection (DCAP)
 - f. CYBER SECURITY

Computable model

David Sanger, author of *The Perfect Weapon*, as quoted on the *Council of Foreign Relations* website

i. Gemini Data vs Palantir

Closing out with debrief

Off to Bar Louie